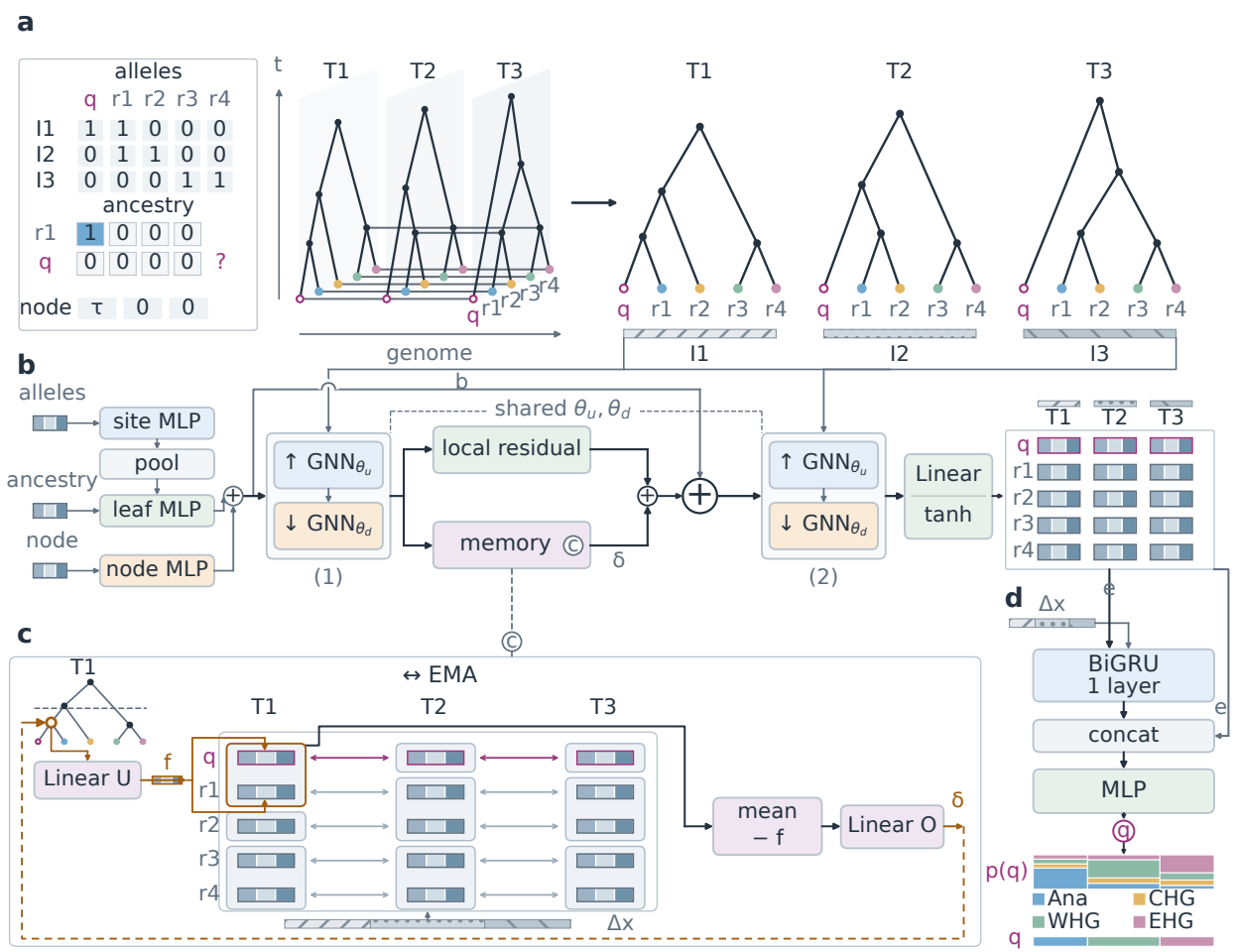

a
alleles
q r1 r2 r3 r4
I1 1 1 0 0 0
I2 0 1 1 0 0
I3 0 0 0 1 1
ancestry
r1 1 0 0 0
q 0 0 0 0 ?
node τ 0 0
t
T1 T2 T3
genome
b
alleles
ancestry
node
site MLP
pool
leaf MLP
node MLP
shared θ_u, θ_d
↑ GNN$_{\theta_u}$
↓ GNN$_{\theta_d}$
local residual
memory ©
δ
(1)
(2)
Linear
tanh
c
↔ EMA
Linear U
f
mean − f
Linear O
Δx
d
Δx
e
BiGRU 1 layer
concat
MLP
@
p(q)
Ana
CHG
WHG
EHG
q